



Forecasting agricultural prices is a critical, yet formidable, challenge.

Timely and reliable price predictions are fundamental to global economic stability. They inform everything from national food policy and risk management to optimising supply chains and mitigating market volatility.

However, these time series are notoriously difficult to forecast due to:

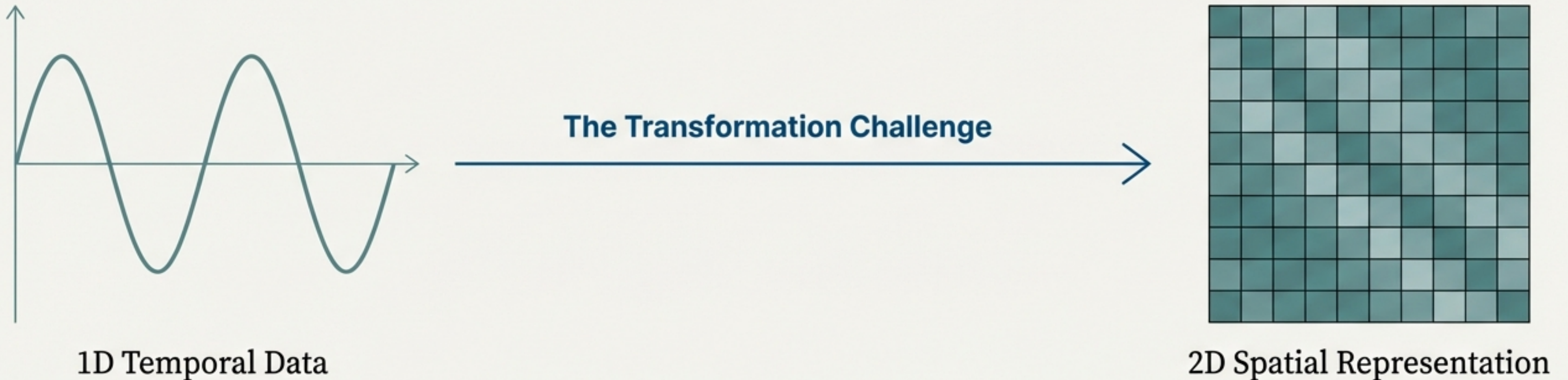
- **Insufficient Data:** Historical data is often limited, making it hard to train robust models.
- **High Volatility & Non-linearity:** Prices are subject to complex, unpredictable factors, from weather events to geopolitical shifts, that defy simple linear analysis.
- The need for **automated and adaptive models** that do not rely on manual feature selection.

How can we enable powerful image recognition models to understand time?

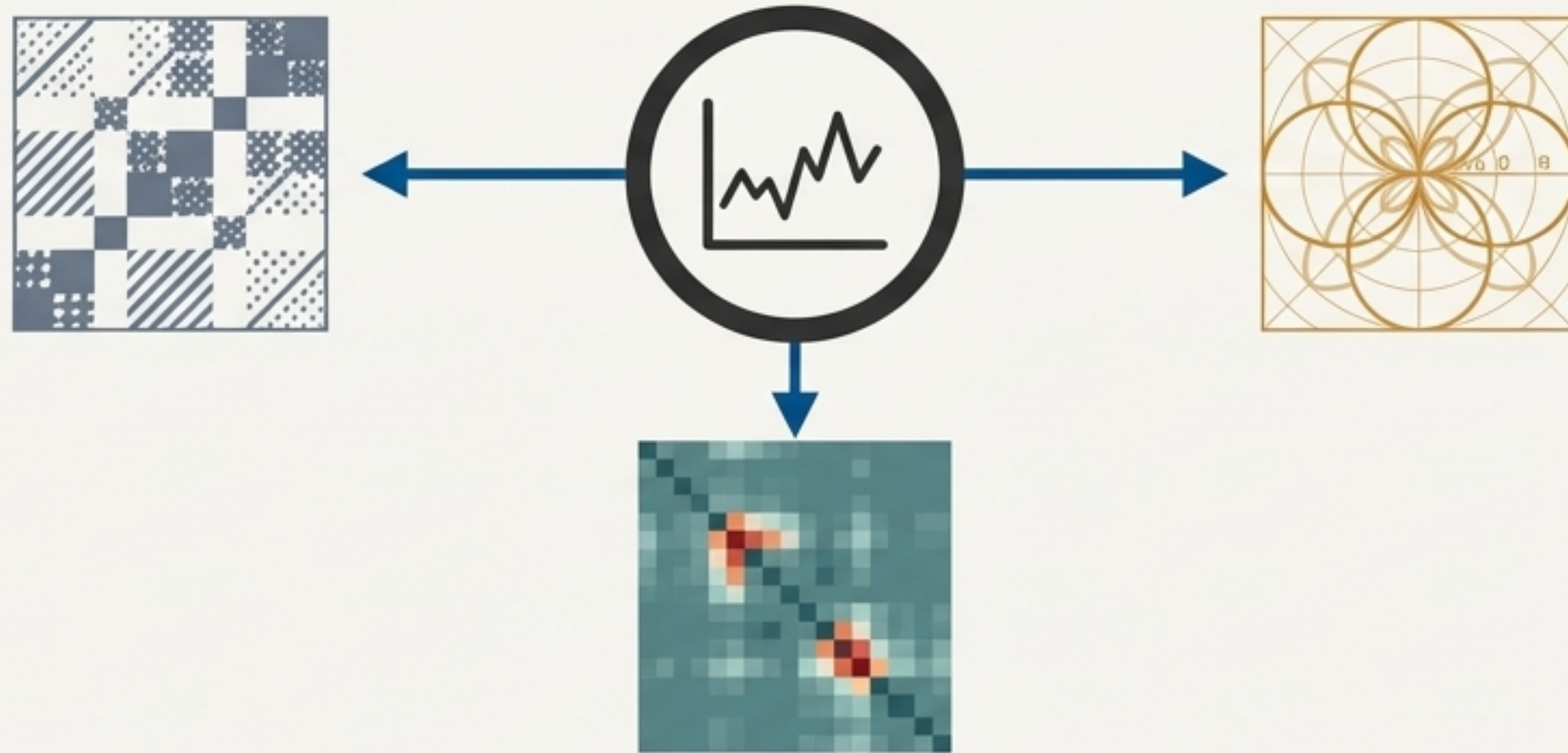
At the heart of the challenge lies a dimensional mismatch.

- Time Series Data is fundamentally one-dimensional. It is a sequence of values over time, capturing temporal dependencies.
- Convolutional Neural Networks (CNNs) are masters of spatial analysis. They excel at identifying intricate patterns, textures, and features within two-dimensional images.

The central question is: How can we translate the temporal dynamics of a time series into a spatial language that a CNN can effectively interpret and learn from?



We can reframe forecasting as a computer vision task.



The proposed approach transforms the one-dimensional time series into three distinct types of two-dimensional images. Each image encoding technique captures a unique aspect of the data's temporal structure, creating a rich, multi-faceted representation. We will generate three 'perspectives' on the data:

1. **A Geometric Fingerprint** (Gramian Angular Field)
2. **A Map of Transitions** (Markov Transition Field)
3. **A History of Recurrence** (Recurrence Plot)

By converting the time series into these images, we can leverage the advanced pattern-recognition capabilities of the advanced pattern-recognition capabilities of state-of-the-art CNNs.

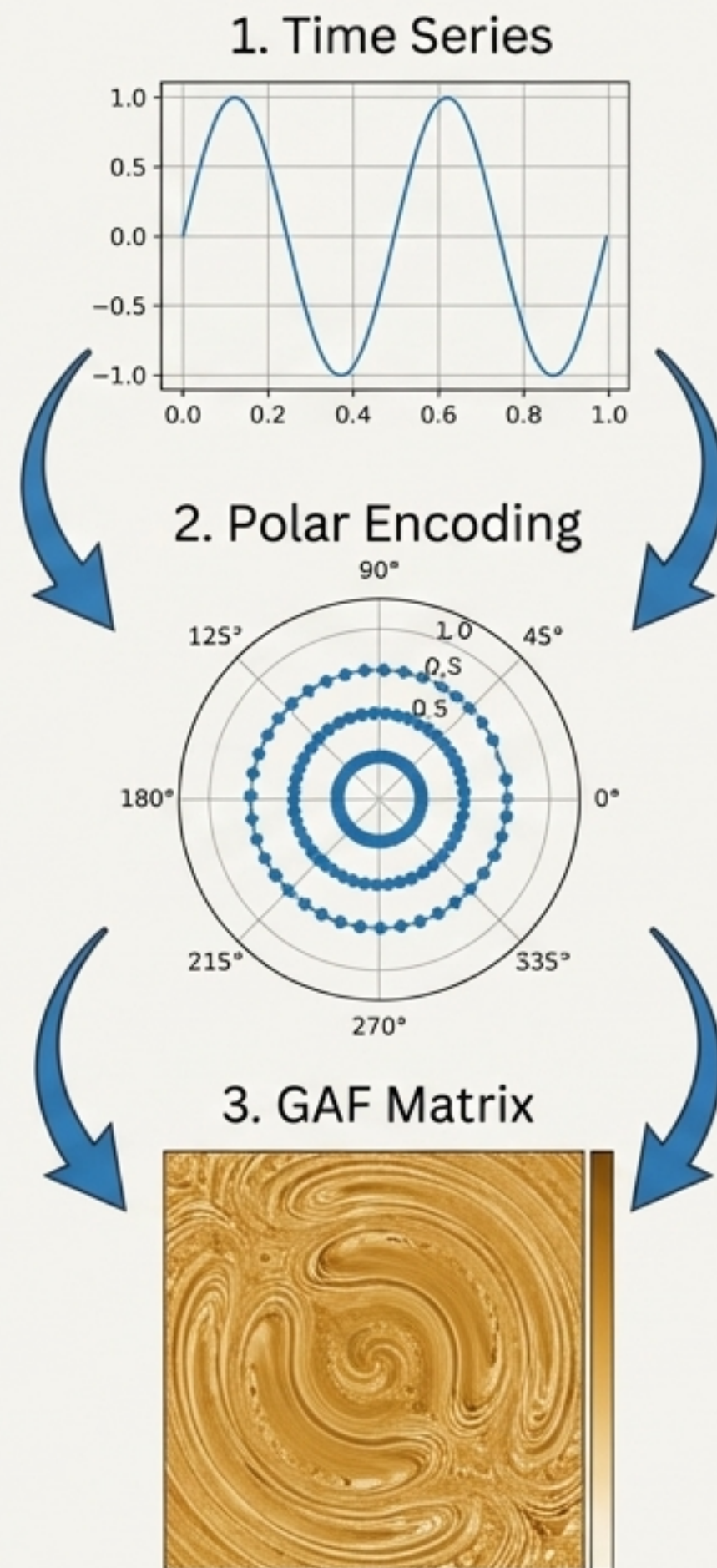
The Geometric Fingerprint: Gramian Angular Fields (GAF)

GAF creates a unique visual texture by preserving temporal correlations from a geometric perspective.

How it works:

1. The time series is first rescaled to a range between $[-1, 1]$.
2. The values are then represented in a polar coordinate system, where the value corresponds to the angle and the time index corresponds to the radius.
3. The final image matrix is constructed from the trigonometric sum/difference of the angles for each pair of points, creating a GAF matrix that is unique to the time series.

This process effectively encodes the series' correlation structure as a geometric 'fingerprint' that a CNN can analyse.



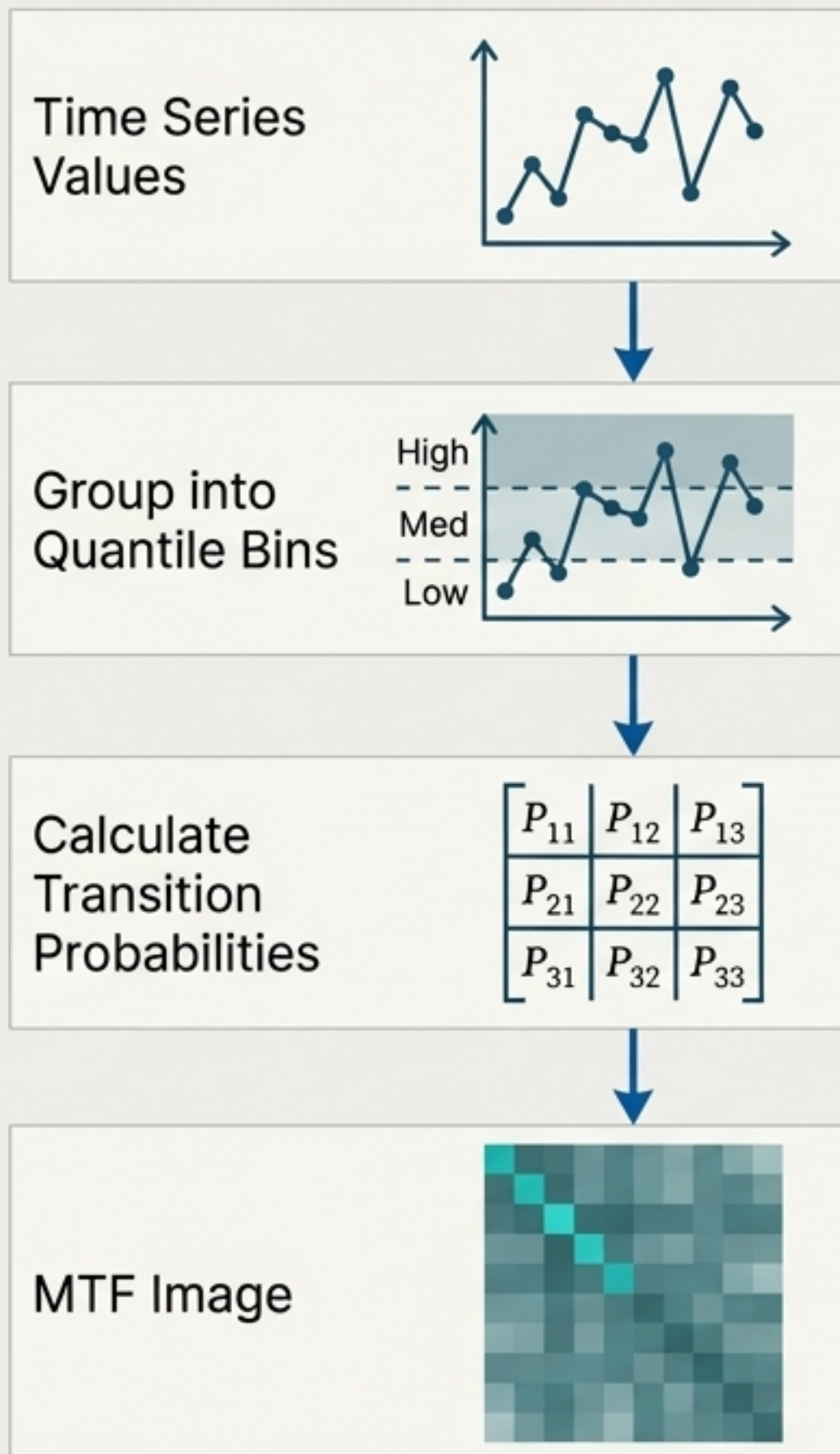
The Map of Transitions: Markov Transition Fields (MTF)

MTF visualises the probability of transitions between different states over time. It acts as a 'choreographer,' mapping the flow of the time series.

How it works:

1. The time series values are discretized into a set of quantile bins (e.g., Q bins).
2. A Markov transition matrix is built, calculating the probability of moving from any bin q_i to any other bin q_j .
3. The final MTF image is a matrix where the pixel at position (i, j) represents the probability of transitioning from the state at time i to the state at time j .

This encodes temporal dependencies as spatial patterns, with trends and periodicities appearing as distinct visual structures in the image.



The History of Recurrence: Recurrence Plots (RP)

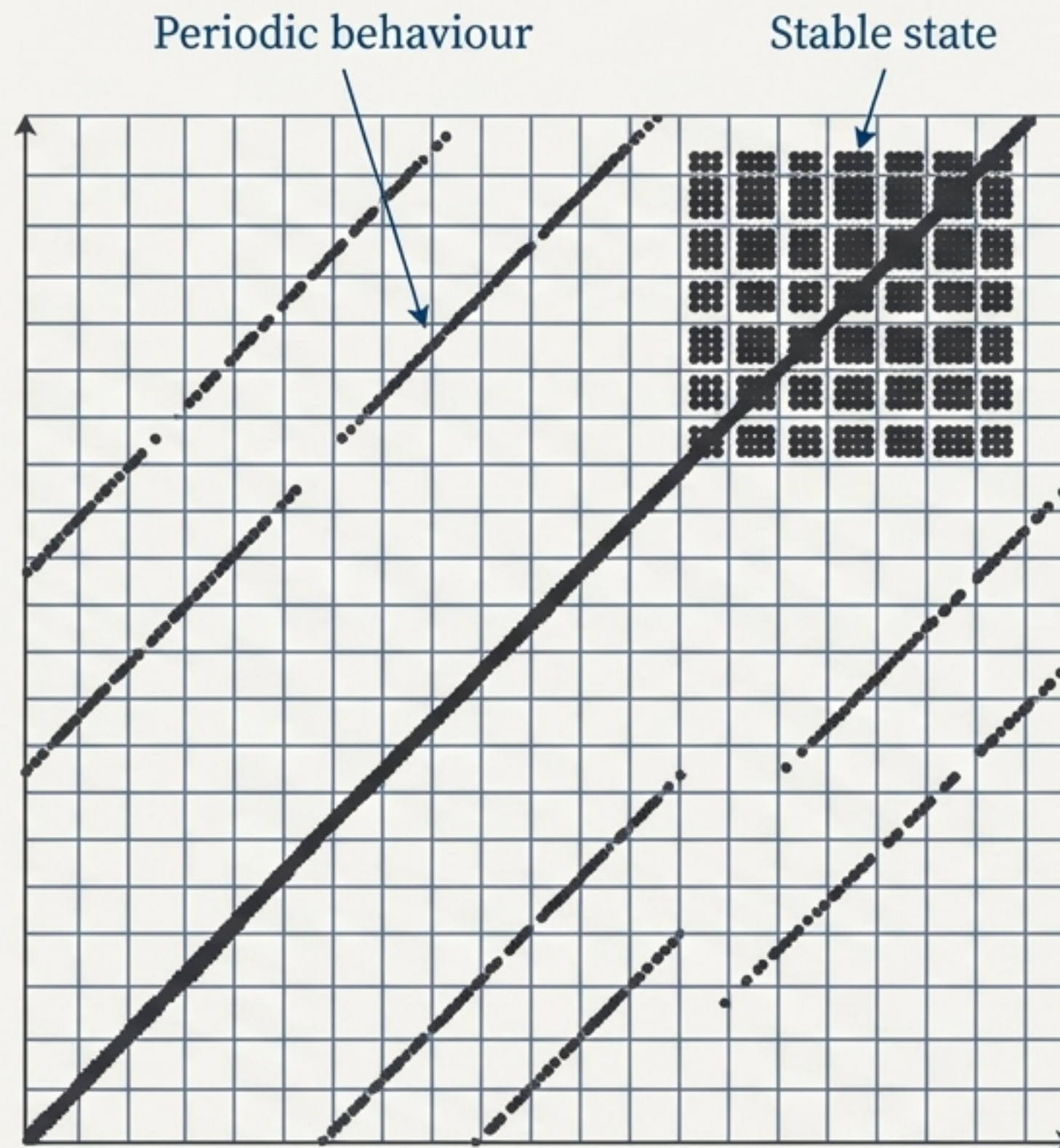
A Recurrence Plot is a visualisation that reveals when a system revisits similar states. It acts as a “historian,” identifying repeating patterns and dynamics.

How it works:

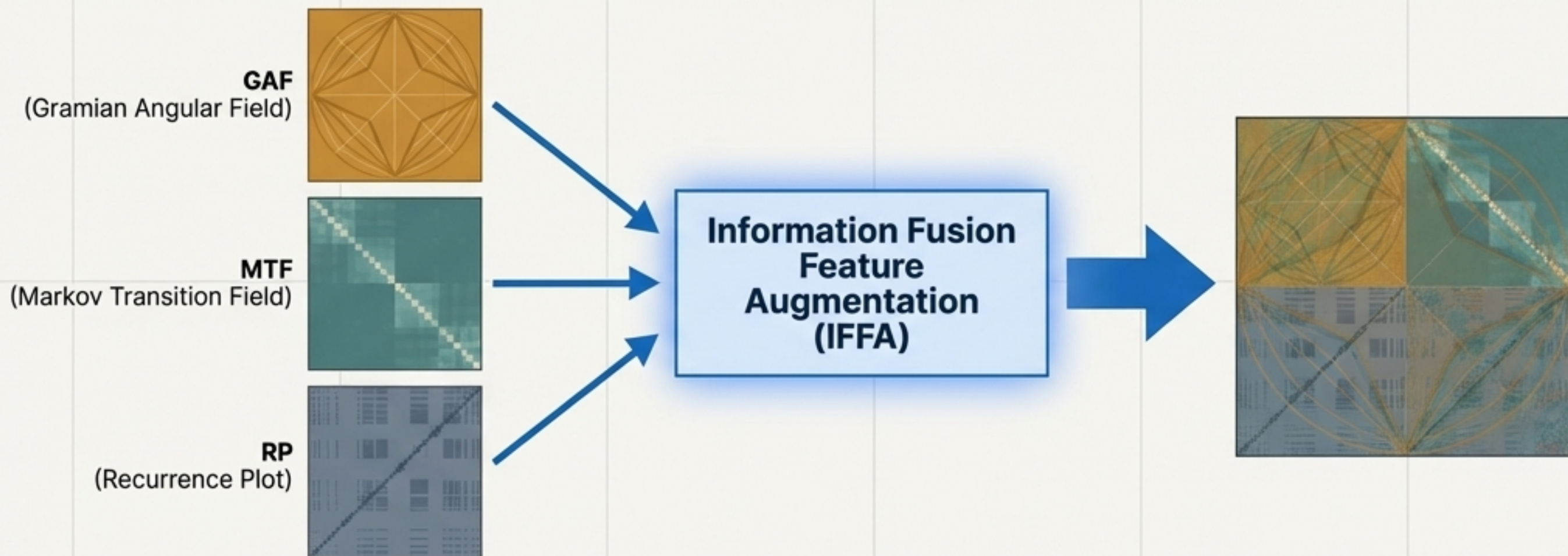
An RP is a square grid where both axes represent time. A point is placed at (i, j) if the state of the system at time i is very close (within a defined threshold) to its state at time j .

The resulting visual patterns are highly informative:

- **Diagonal Lines:** Indicate deterministic or periodic behaviour.
- **Vertical/Horizontal Lines:** Show states that change very little.
- **Large Blocks:** Indicate the system remaining in a similar state for an extended period.



The Innovation: Fusing Multiple Perspectives into a Single, Richer View



While each encoding provides a valuable perspective, the true power lies in combining them. **Information Fusion Feature Augmentation (IFFA)** is a novel technique that integrates the features from all three images.

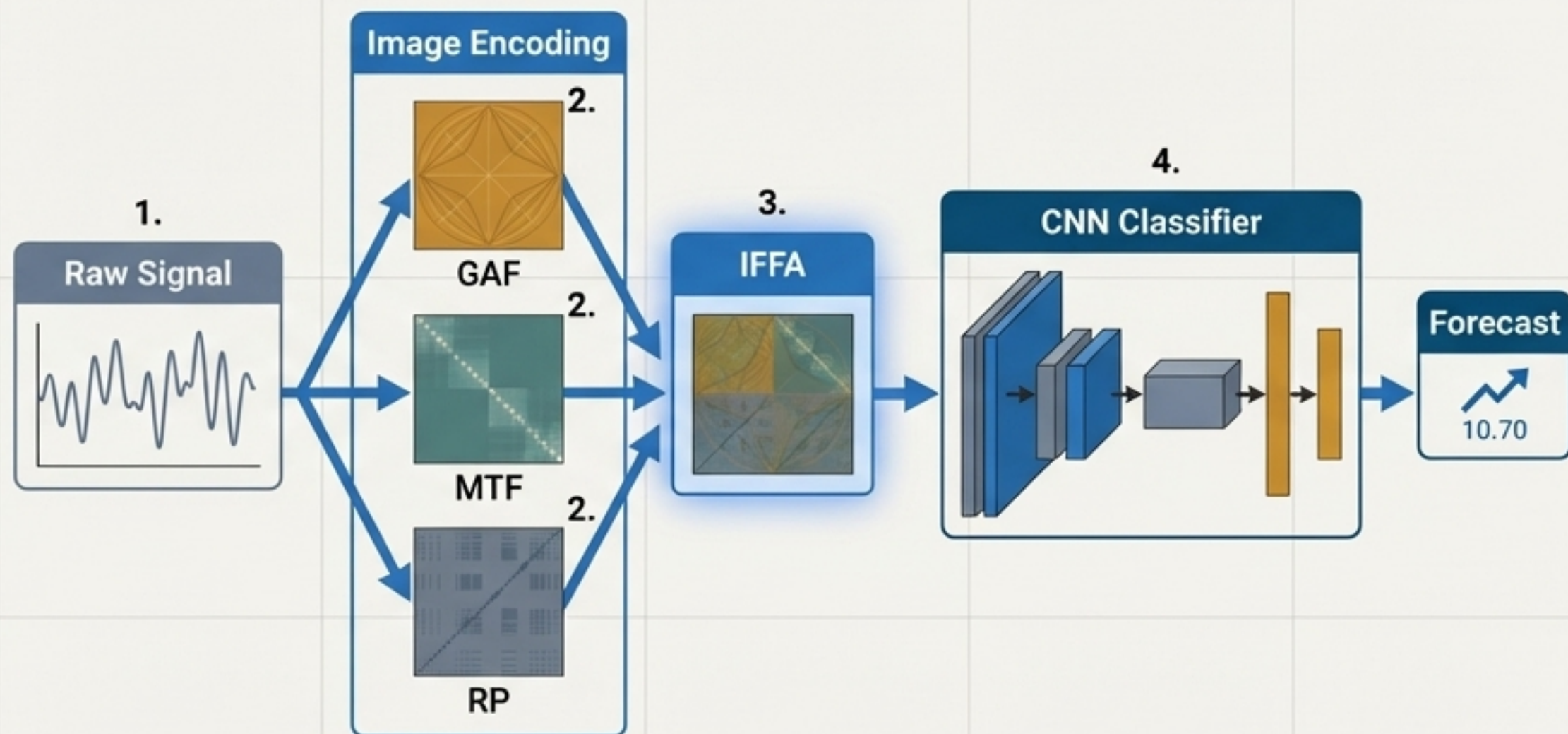
This process creates a single, multi-channel “super-image” that contains the complementary information from the GAF, MTF, and RP representations. By fusing these features, the model gains a more comprehensive and robust understanding of the time series’ dynamics than any single view could provide.

The result is an information-rich input that is greater than the sum of its parts.

The IFFA-TSCI-CNN Pipeline: From Raw Signal to Forecast

This diagram shows the complete workflow for the proposed forecasting model selection method.

1. **Time Series Preprocessing:** The raw agricultural price data is segmented and prepared.
2. **Time Series Image Encoding:** The three encoding methods (GAF, MTF, RP) are applied to generate the image representations.
3. **Information Fusion Feature Augmentation (IFFA):** The features from the three images are fused into a single, combined image.
4. **CNN Classifier Training:** A Convolutional Neural Network (such as ResNet-18, DenseNet-121, or VGG-11) is trained on these combined images to classify and select the optimal forecasting model for a given time segment.



Evaluating Performance on Real-World Agricultural Data

The methodology was rigorously tested on two key agricultural futures price datasets.

Datasets



Corn: Daily prices from January 1989 to December 2019 (10,950 data points).

Wheat: Daily prices from January 1999 to December 2019 (7,300 data points).

Evaluation Metric

$$\text{MAPE} = \frac{\sum |\text{act} - \text{prd}|}{a} \times \frac{100}{n}$$

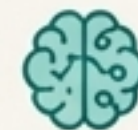
Mean Absolute Percentage Error (MAPE): A standard measure of prediction accuracy.

A lower MAPE score indicates a better-performing model.

Models Compared



Forecasting Models



ML Models



DL Models

Forecasting Models: ARIMA, Theta, RW, BP, ELM, ETS

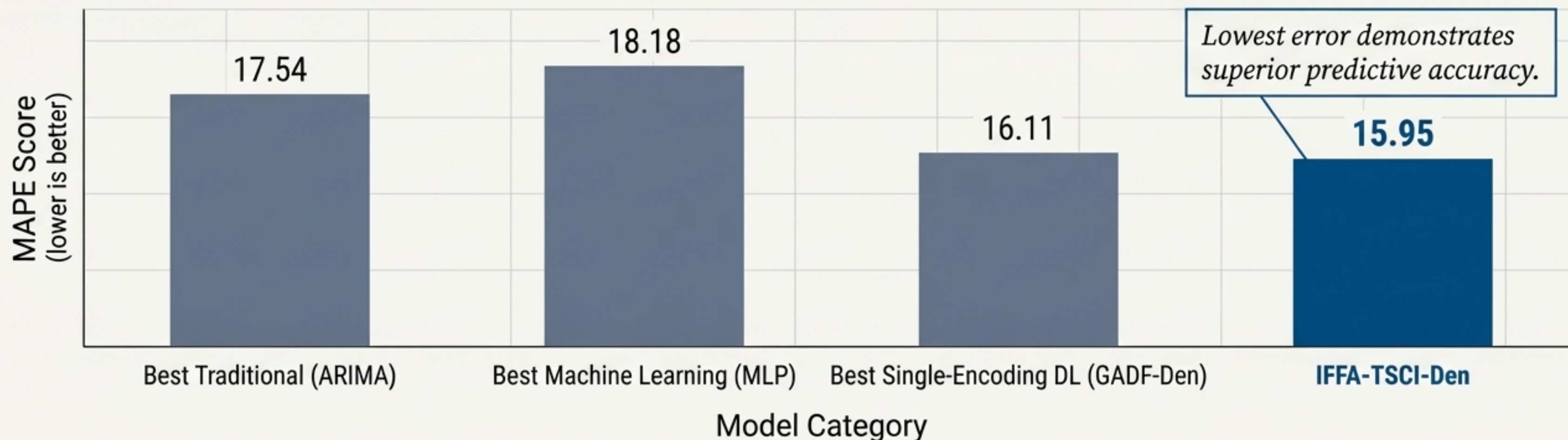
Machine Learning Models: SVM, DTW, MLP

Deep Learning Models: CNNs (ResNet, DenseNet, VGG) using single image encodings vs. the proposed IFFA fusion method.

The IFFA-TSCI method consistently outperforms established approaches.

Across both datasets and experimental schemes, the proposed method of fusing image features and using a deep learning classifier delivered superior forecasting accuracy. The model based on the fused “super-image” consistently achieved the lowest prediction error.

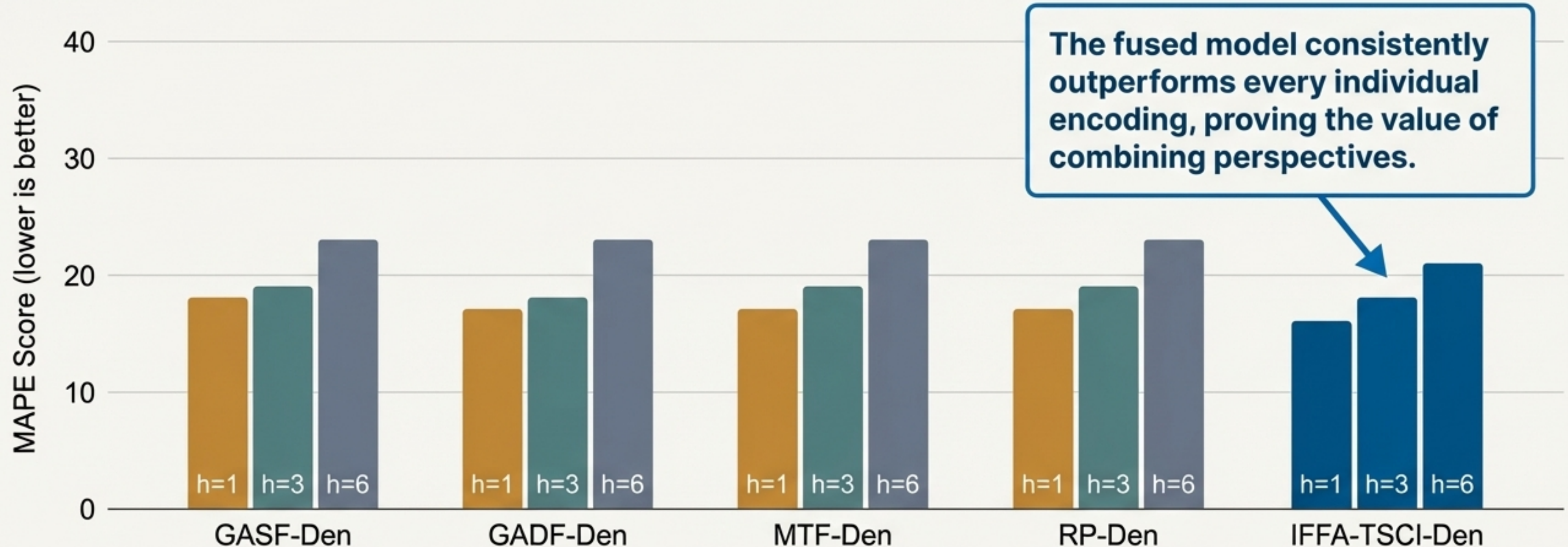
MAPE Comparison (Prediction Horizon $h=3$)



Feature fusion is the key driver of improved performance.

Comparing the fused IFFA model against models trained on single-encoding images reveals the significant benefit of the fusion strategy. The combined image provides the CNN with a richer feature set, leading to a consistent reduction in forecasting error across all network architectures.

MAPE of DenseNet-121 Models (S-FCV Scheme)



Superior performance is demonstrated across multiple experimental schemes.

The method's effectiveness was validated using two distinct experimental approaches to handle the limited data:

- **Transfer Learning (TL):** Mitigates the issue of insufficient training data.
- **S-Folder Cross Validation (S-FCV):** A robust validation method for time series.

As shown in the results, the IFFA-TSCI-CNN model consistently outperforms both traditional and single-encoding deep learning models within both the TL and S-FCV frameworks, confirming the robustness and reliability of the approach.

ResNet-18, h=1 Comparison

Transfer Learning (TL) Scheme

IFFA-TSCI-Res MAPE: **14.68** ✓

GADF-Res MAPE: 14.77

S-Folder Cross Validation (S-FCV) Scheme

IFFA-TSCI-Res MAPE: **13.64** ✓

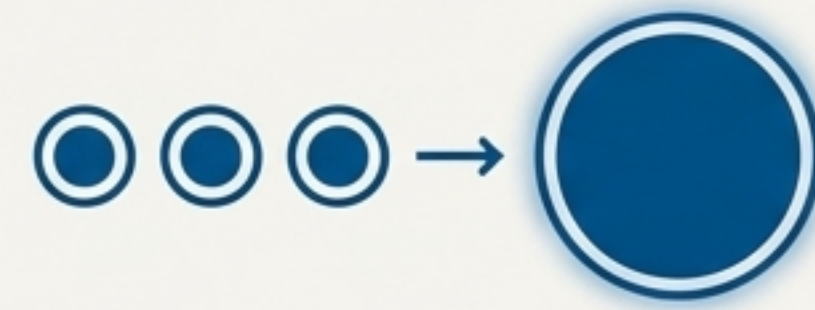
GADF-Res MAPE: 14.21

Key Contributions: A New Framework for Time Series Analysis



1. Transformation of Predictive Modelling into Time Series Image Classification

By redefining the problem, we unlock the ability to use powerful, state-of-the-art computer vision techniques for forecasting, moving beyond traditional feature engineering.



2. Enhancement of Features via Information Fusion (IFFA)

The novel fusion method creates a more descriptive and robust input for the CNN. This allows the model to capture more meaningful patterns within the data, leading to a significant improvement in overall security and predictive accuracy.

Beyond Agriculture: A Generalisable Method for Complex Sequential Data



Finance:

Forecasting stock prices and detecting market volatility



Engineering & IoT:

Predictive maintenance and monitoring of sensor data from industrial machinery



Time Series



Fused Image



Insight

While demonstrated on agricultural prices, the principle of transforming temporal dynamics into fused spatial representations is a powerful and widely applicable technique. This approach provides a new foundation for feature research into multimodal and macro-micro time series series prediction tasks.



Healthcare:

Analysing physiological signals like ECG or EEG for anomaly detection



Climatology:

Modelling and predicting complex weather patterns